MILITARY SPECIFICATION

CLOTH, TWILL, POLYESTER, 6.0 OUNCE (TEXTURIZED WARP-SPUN FILLING)

This specification is approved for use by Naval Supply Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers one type of twill cloth made from 100% texturized polyester.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

Beneficial comments recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760, by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580).

2.2 Other publications The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply.

Technical Manual of the American Association of Textile Chemists and Colorists

(Application for copies should be addressed to AATCC National Headquarters, Research Triangle Park, PO Box 12215, Durham, NC).

3. REQUIREMENTS

3.1 Standard sample The finished cloth shall match the standard and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article When specified, the contractor shall furnish a sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Materials

3.3.1 Fiber The fiber shall be polyethylene glycol terephthalate, regular tenacity, semi-dull (see 4.4.1).

3.3.2 Yarn The yarn for the warp shall be nominal 150 denier, octalobal cross section, 34 $\pm$ 3 filaments, polyester fiber and shall be texturized by the false twist process. All yarns shall contain the same number of filaments. The filling yarn shall be composed of a nominal 3 denier, 2 inch (5.1cm) staple, pill-resistant polyester fiber.

3.4 Color The color of the finished cloth shall be blue 3346 and shall match the standard sample.

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3.4.1 Matching The color of the finished cloth shall match the standard sample under artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.4.2 Colorfastness The finished cloth shall show fastness to light, laundering, wet-dry cleaning, perspiration and crocking equal to or better than the standard sample. When no standard sample is available, the finished dyed cloth shall show a minimum of "good" fastness to light, wet-dry cleaning, and perspiration, and shall show a Munsell Value for crocking not lower than 8.5. Testing shall be as specified in 4.5.

3.5 Physical requirements The finished cloth shall conform to the requirements specified in Table I when tested as specified in 4.5.

Table I - Physical requirements

Weight oz/sq yd (g/m^2)	Yarns per inch (cm) (Min)		Breaking strength lbs (newtons) (min)		Tearing strength lbs (newton) (min)	
	Warp	Filling	Warp	Filling	Warp	Filling
6.0 ± 0.5 (203 ± 17)	66 (26.0)	58 (22.9)	225 (1001)	70 (312)	10 (46)	7 (31)

3.5.1 Weave The weave shall be a $\frac{2}{1}$ right hand twill.

3.5.2 Width Unless otherwise specified (see 6.2) the width of the cloth shall be a minimum of 60 inches (152.4cm).

3.6 Finish The cloth shall be scoured, heat set and dyed. The character of the finish shall be equal to that of the standard sample.

3.7 Non-fibrous materials The starch and protein content including chloroform-soluble and water-soluble material of the cloth shall not exceed 3.0 percent when tested as specified in 4.5.

3.8 Dimensional stability The finished cloth shall not shrink or elongate more than 2.5 percent in the warp and filling when tested as specified in 4.5.

3.9 pH The pH value of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.5.

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3.10 Length and put-up Unless otherwise specified (see 6.2), the finished cloth shall be furnished in continuous lengths each not less than 40 yards (36.6m). Each length shall be put up on rolls as specified in PPP-P-1133.

3.11 Fiber identification Each roll shall be labeled or ticketed for fiber content in accordance with the Textile Fibers Products Identification Act.

3.12 Face identification The face side of the cloth shall be identified by stamping on that side with the word "Face" at each end of the roll.

3.13 Workmanship The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards ($83.6m^2$) when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

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4.3 First article inspection When required, the first article of finished cloth shall be visually inspected for color and finished appearance and shall be tested for chemical and physical properties in accordance with the applicable methods specified in 4.5.

4.4 Quality conformance inspection Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified, in this specification or applicable purchase document. In addition, a certificate of compliance shall be submitted for the material requirements of 3.3.1 and 3.3.2.

4.4.2 Examination of the end item Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.4.

4.4.2.1 Yard-by-yard examination Each roll in the sample shall be examined on face side only. When the total yardage in the roll does not exceed 100 yards (91.4m), the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.4m) only 100 yards (91.4m) shall be examined. All defects as defined in Section III of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) (0.91m) shall be scored and assigned demerit points as listed in 4.4.2.1.1, except that only those knots which exceed the limits shown on Sears Fabric Defect Scale (see 6.4), "C" knots shall be scored. No linear yard (increments of 1 yard (0.91m) on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of the total yardage examined exceeds 25 points. The lot shall be unacceptable if the point per 100 square yards ($83.6m^2$) of two or more individual rolls exceeds 38 points. If one roll exceeds the point level per 100 square yards ($83.6m^2$), a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 38 points per 100 square yards ($83.6m^2$). Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample X 3600}}{\text{Contracted width of cloth (inches) X total yards inspected}} = \text{Points per 100 square yards}^2 \text{ (83.6m}^2\text{)}$$

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4.4.2.1.1 Demerit points Demerit points shall be assigned as follows:

For defects 3 inches (7.6cm) or less in any dimensions - one point
For defects exceeding 3 inches (7.6cm), but not exceeding
6 inches (15.2cm) in any direction - two points
For defects exceeding 6 inches (15.2cm), but not exceeding
9 inches (22.9cm) in any dimension - three points
For defects exceeding 9 inches (22.9cm) in any dimension - four points

The following defects, when present, shall be scored 4 points for each yard (0.9m) in which they occur:

Baggy, ridgy or wavy cloth
Width less than specified
Characteristics of finish, hand and shade not equal to the standard sample
Non uniformity of shade (poor dye penetration, mottled, streaky or cloudy)
Holes, cuts, tears, open places

4.4.2.2 Examination for length

4.4.2.2.1 Individual rolls During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards (1.82m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective with respect to length.

4.4.2.2.2 Total yardage in sample The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.4.2.3 Examination for shade variation During the yard-by-yard examination, each roll in the sample shall be examined for shade variation. Any role in the sample exhibiting uneven shade, shade variation side to side, side to center, or end to end, shall be cause for rejection of the entire lot represented by the sample.

4.4.2.4 Examination for shade match and finish, individual rolls Each roll in the lot shall be examined visually for shade match (see 3.4.2) and finish (see 3.6). A roll shall be unacceptable if it fails to meet the requirements for shade match or finish. The sample unit shall be a 4 inch by 20 inch (10.2 by 50.8cm) swatch of the cloth. A sample unit shall be drawn from each roll in the lot.

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4.4.2.5 Examination for face identification and non-conformance with the Textile Fiber Products Identification Act During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.4.3 Examination of packaging requirements An examination shall be made in accordance with the provision of PPP-P-1133 to determine that packaging, packing and marking comply with Section 5 requirements of this specification.

4.5 Testing of the end item The methods of testing specified in FED-STD-191, wherever applicable and as listed in Table II shall be followed. The physical and chemical values specified in Section 3, except where otherwise indicated, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 2 yards (1.82m), full width of the finished cloth. The lot size shall be expressed in units of one yard (0.91m). The sample size shall be as follows and the lot shall be unacceptable if one or more sample units fail to meet any requirements specified.

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table II - Test Methods

Characteristic	Requirement Paragraph	Test Method
Colorfastness to:		
Light	3.4.2	5660
Laundering	3.4.2	5614 1/
Crocking	3.4.2	5651
Perspiration	3.4.2	5680
Wet-dry cleaning	3.4.2	5622
Weight	3.5	5041
Yarns per inch:		
Warp	3.5	5050
Filling	3.5	5050
Breaking strength:		
Warp	3.5	5100
Filling	3.5	5100
Tearing strength:		
Warp	3.5	5132
Filling	3.5	5132

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Table II - Instructions for Testing (cont'd)

Characteristic	Requirement Paragraph	Test Method
Dimensional stability		
Warp	3.8	2/
Filling	3.8	2/
Non-fibrous material content	3.7	2611
pH	3.11	2811

1/ Only the stain on the polyester fibers of the transfer cloth shall be evaluated.

2/ AATCC Test Method 135-1973, Dimensional Changes in Automatic Home Laundering of Durable Press Woven or Knit Fabrics - Method II (120 ± 5 F), Drying procedure B (tumble). Specimen shall be given 5 cycles before testing.

5. PACKAGING

5.1 Preservation-packaging Packaging shall be level A or C (see 6.2).

5.1.1 Levels A and C The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing Packing shall be level A, B or C (see 6.2).

5.2.1 Levels A, B and C The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1133.

6. NOTES

6.1 Intended use The cloth covered by this specification is intended for use in the manufacturing of woman's winter shirts.

* 6.2 Ordering data Procurement documents should specify the following:

a. Title, number and date of this specification.
b. When first article is required (see 3.2).
c. Minimum width of cloth, if other than specified (see 3.5.2).
d. Minimum length, if other than specified (see 3.10).
e. Levels of packaging and packing requirements (see 5.1 and 5.2).

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6.3 Standard sample For access to the standard sample, address the procuring office issuing the invitation for bids.

6.4 Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 733Q "Sears Defect Replica Kit", Sears Tower, Chicago, ILL 60684.

Custodian:
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Preparing Activity:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-87051(SA) CLOTH, TWILL, POLYESTER, 6.0 OUNCE

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